AWS Monitoring & Audit: CloudWatch, CloudTrail & Config

Exam: AWS Certified Solutions Architect - Associate (SAA-C03)

1. AWS Monitoring - Section Introduction

To maintain operational excellence and performance efficiency (two pillars of the Well-Architected Framework), you must understand how AWS resources are performing and who is accessing them.

- **CloudWatch:** "What is happening right now?" (Performance & Logs)
- **CloudTrail:** "Who did what?" (Audit & API History)
- **Config:** "What does the resource look like?" (Configuration & Compliance)

2. CloudWatch Metrics

CloudWatch provides data and actionable insights to monitor your applications, respond to system-wide performance changes, optimize resource utilization, and get a unified view of operational health.

- **Metrics:** A variable to monitor (e.g., `CPUUtilization`).
- **Namespaces:** A container for CloudWatch metrics (e.g., `AWS/EC2`).
- **Dimensions:** A name/value pair that helps you uniquely identify a metric (e.g., `InstanceId`).
- **Timestamps:** All metrics have a timestamp.
- **Metric Resolution:**
  - **Standard Resolution:** 1-minute granularity (default for custom metrics).
  - **High Resolution:** 1-second granularity (cost is higher).
- **Dashboards:** Global access (can view dashboards from different regions).

Exam Tip:

- By default, EC2 metrics are sent every **5 minutes** (basic monitoring). You can enable **Detailed Monitoring** to get data every **1 minute** (costs extra).
- **RAM/Memory usage is NOT a default metric** for EC2. You must push this as a *Custom Metric*.

3. CloudWatch Logs

Centralized log management. You can store, access, and monitor logs from EC2 instances, AWS CloudTrail, Route 53, and other sources.

- **Log Groups:** Usually represent an application or a specific source (e.g., `/aws/lambda/my-function`).
- **Log Streams:** Instances within a specific application or log files (e.g., a specific container instance log).
- **Metric Filters:** You can search logs for a specific phrase (e.g., "Error" or "404") and turn that into a numerical metric to trigger alarms.
- **Export:** Logs can be exported to **S3** (for long-term archival) or streamed to **Kinesis/Lambda/OpenSearch** (for real-time analysis).

Exam Tip:

- Logs in S3 can take up to 12 hours to become available for export. For real-time processing, use **Subscription Filters** to send logs to Kinesis or Lambda.
- Logs never expire by default (indefinite retention). You must set a retention policy to delete them automatically.

4. CloudWatch Logs - Live Tail

- Provides an interactive, scrolling view of new log events as they are ingested.
- Similar to the `tail -f` command in Linux.
- Helps dev/ops teams debug issues quickly in real-time without needing to constantly refresh the console or run queries.

5. CloudWatch Agent & CloudWatch Logs Agent

To get logs or system-level metrics (like disk space or RAM) from inside an EC2 instance, you need an agent.

- **CloudWatch Logs Agent (Old):** Only sent logs. Deprecated.
- **Unified CloudWatch Agent (New):**
  - Collects **both** logs and system-level metrics (RAM, Swap, Disk Space, Netstat, etc.).
  - Works on EC2 and on-premises servers.
  - Configuration is stored in the Parameter Store (SSM) for easy deployment across fleets.

Exam Tip:

- If a question asks how to monitor "Memory Utilization" or "Disk Space Available" on EC2, the answer is always **CloudWatch Agent** (Push Custom Metric).

6. CloudWatch Alarms

Alarms are used to trigger notifications or actions based on metric thresholds.

- **Three States:**

1. **OK:** Metric is within the threshold.
2. **ALARM:** Metric crosses the threshold.
3. **INSUFFICIENT_DATA:** Not enough data (e.g., new instance, connectivity issue).

- **Actions (Targets):**
  - **SNS:** Send an email/SMS.
  - **Auto Scaling:** Scale out or Scale in (ASG).
  - **EC2 Actions:** Stop, Terminate, Reboot, or **Recover** an instance.

Exam Tip:

- **EC2 Instance Recovery:** If an instance fails a *System Status Check* (hardware issue), a CloudWatch Alarm can automatically "Recover" the instance. This moves the instance to new hardware but keeps the Instance ID, Private IP, and Elastic IP the same.

7. CloudWatch Synthetics (Canaries)

- **What it is:** Configurable scripts (Node.js or Python) that run on a schedule to monitor your endpoints and APIs.
- **Function:** It simulates a user navigating your application ("Synthetic traffic").
- **Use Case:** Verify your application is up, check latency, and ensure UI elements (like a checkout button) are working even when there is no real user traffic.

8. EventBridge Overview (formerly CloudWatch Events)

EventBridge is a serverless event bus that connects applications using data from your own apps, SaaS apps, and AWS services.

- **Evolution:** CloudWatch Events is being replaced by EventBridge (same underlying technology, more features).
- **Structure:**
  - **Source:** The service generating the event (e.g., "EC2 Instance State Change").
  - **Rules:** Filter incoming events (e.g., "Only look for EC2 stopped events").
  - **Target:** The service to trigger (Lambda, SNS, SQS, Kinesis, Step Functions).
- **Schema Registry:** Allows you to discover, create, and manage schemas for events.
- **Scheduled Events:** Functions like a "Cron Job" (e.g., Trigger a Lambda function every night at 2 AM).

9. CloudWatch Insights and Operational Visibility

Advanced features for deeper analysis.

- **Container Insights:** Collects metrics/logs from containerized applications (ECS, EKS, Fargate, Kubernetes).
- **Lambda Insights:** Monitoring and troubleshooting for serverless functions (cold starts,

memory usage).

- **Contributor Insights:** Analyzes log data to find "Top N" contributors (e.g., "Who are the top 10 IP addresses hitting my load balancer?").
- **Application Insights:** Detects errors/problems in .NET and SQL Server databases (and other stacks) using AI/ML.

10. CloudTrail Overview

Governance, compliance, and operational and risk auditing. CloudTrail records **API calls** made within your AWS account.

- **What it records:** Who (User/Role), When (Time), What (API Action), Where (Resource & Source IP).
- **Default:** Enabled by default. Keeps 90 days of history in the console (Management Events).
- **Types of Events:**
  1. **Management Events:** Operations on the AWS account (e.g., `AttachRolePolicy`, `CreateVPC`, `ConsoleLogin`). Enabled by default.
  2. **Data Events:** Operations within a resource (e.g., S3 object upload, Lambda function invocation, DynamoDB item put). **High volume, disabled by default.**
  3. **Insights Events:** Detects unusual activity (spikes in resource provisioning).
- **Storage:** To keep logs longer than 90 days, you must configure a **Trail** to send logs to S3.

Exam Tip:

- If you need to know "Who terminated the EC2 instance?" or "Who deleted the S3 bucket?", the answer is **CloudTrail**.

11. CloudTrail - EventBridge Integration

CloudTrail allows you to *react* to API calls. CloudTrail logs the API call -> Sends event to EventBridge -> EventBridge triggers a target.

- **Use Case:** Security automation.
  - *Scenario:* A user creates an S3 bucket that is Public.
  - *Flow:* CloudTrail records `CreateBucket` -> EventBridge Rule matches pattern -> Triggers Lambda -> Lambda deletes the bucket or fixes the policy.

12. AWS Config - Overview

Detailed view of the configuration of AWS resources in your AWS account.

- **Inventory:** Records resource configurations and changes over time (Versioning for infrastructure).

- **Compliance:** You can set "Config Rules" (Managed or Custom).
  - *Example:* "Is SSH (port 22) unrestricted?" "Are EBS volumes encrypted?"
- **Remediation:** Can automatically fix non-compliant resources using SSM Automation Documents.
- **Notifications:** Sends SNS notifications when a resource violates a rule.

Exam Tip:

- If you need to view the **history of configuration changes** (e.g., "What did the Security Group rules look like yesterday?"), use **AWS Config**.
- It is *Regional* service.

13. Summary: CloudWatch vs. CloudTrail vs. Config

Feature	CloudWatch	CloudTrail	AWS Config
Primary Question	"What is the performance?"	"Who made the call?"	"How is it configured?"
Focus	Performance, Metrics, Logs, Alarms	Auditing, Security, API History	Compliance, Inventory, Change History
Key Use Case	CPU is high; App is throwing 500 errors.	Who deleted the production DB?	Is my S3 bucket public? Did the SG change?
Real-time action	Alarms & EventBridge	EventBridge Integration	Config Rules & Remediation
Retention	Metrics: 15 months; Logs: Indefinite	90 days (Console); S3 (Indefinite)	Indefinite (based on settings)

Cheat Sheet Mnemonic:

- **Watch** = Performance (Speed, Errors, Load)
- **Trail** = Audit (Spy on users, API calls)
- **Config** = Rules (Policy, Settings, History)